Nptel Online Certification Course
Indian Institute of Technology Kharagpur
Computer Vision
Assignment - Week 12

Number of questions: 10 **Total marks: 10x2=20**

QUESTION 1: **Type: MSQ**

Which of the following statements are true for Pooling?

a) It accumulates training data into a central buffer to optimize the gradient descent optimization process.

b) It provides translational invariance, allowing the network to detect features even if they are shifted, rotated, or slightly distorted.

c) It reduces the spatial dimensions (width and height) of the feature maps, thereby decreasing the number of parameters and computational complexity.

d) It selectively extracts dominant features (like maximum or average values) while discarding less prominent activations.

Correct Answer: b), c) and d)
Detailed Solution:
a) False. Pooling has nothing to do with pooling training data together. It operates strictly on the internal feature maps (activations) layer-by-layer within the network.

QUESTION 2: **Type: Numeric**

Up-convolve the following input x using the filter h. What is the value at (2,1) in the up-convolved feature map? We assume index of first cell of any 2D-matrix is (0,0).

x

0	3
3	2

h

0	0.1	0
0.1	0.2	0.1
0	0.1	0

Correct Answer: 0.8

Detailed Solution:

(0×0)	$(0 \times 0.1) + (3 \times 0)$	$(0 \times 0) + (3 \times 0.1)$	(3×0)
$(0 \times 0.1) + (3 \times 0)$	$(0 \times 0.2) + (3 \times 0.1)$ $+(3 \times 0.1) + (2 \times 0)$	$(0 \times 0.1) + (3 \times 0.2)$ $+(3 \times 0) + (2 \times 0.1)$	$(3 \times 0.1) + (2 \times 0)$
$(0 \times 0) + (3 \times 0.1)$	$(0 \times 0.1) + (3 \times 0)$ $+(3 \times 0.2) + (2 \times 0.1)$	$(0 \times 0) + (3 \times 0.1)$ $+(3 \times 0.1) + (2 \times 0.2)$	$(3 \times 0) + (2 \times 0.1)$
(3×0)	$(3 \times 0.1) + (2 \times 0)$	$(3 \times 0) + (2 \times 0.1)$	(2×0)

0	0 + 0	0 + 0.3	0
0 + 0	0 + 0.3 + 0.3 + 0	0 + 0.6 + 0 + 0.2	0.3 + 0
0 + 0.3	0 + 0 + 0.6 + 0.2	0 + 0.3 + 0.3 + 0.4	0 + 0.2
0	0.3 + 0	0 + 0.2	0

0	0	0.3	0
0	0.6	0.8	0.3
0.3	**0.8**	1.0	0.2
0	0.3	0.2	0

QUESTION 3: **Type: Numeric**

An autonomous driving system's object recognition model classifies a detected obstacle into one of four classes: Pedestrian, Bicycle, Car, or Truck. For a specific sample, the model outputs the softmax predictions $[0.08, 0.12, 0.65, 0.15]$. Calculate the cross-entropy loss for this sample (using the natural logarithm) if the actual ground truth object is a Truck, represented by the true labels $[0, 0, 0, 1]$. Round off the answer to 4 places of decimals.

Correct Answer: 1.8971

Detailed Solution: $L = -\sum y \cdot \log(p)$ where y is the true label and p is the predicted probability. In this case, we calculate it as: $L = -[1 \cdot \log(0.15)] = 1.8971$

QUESTION 4: **Type: MSQ**

Which of the following statements are true for Batch normalisation?

a) It returns the normalised mean and standard deviation of the weights.

b) It normalizes the intermediate layer activations across a mini-batch during training.

c) It is a very efficient backpropagation technique.

d) It allows using higher learning rate.

Correct Answer: b), d)

Detailed Solution:

Batch normalization calculates the mean and variance for each mini-batch to stabilize the feature distributions and mitigate internal covariate shift. By preventing gradients from exploding or vanishing across layers, it safely allows the network to train with much larger learning rates.

FOR QUESTIONS 5, 6 AND 7:

Consider an input image of dimension $27 \times 21 \times 3$. It passes through a convolutional layer with 32 kernels of dimension $3 \times 3 \times 3$. The image has been zero padded on both sides with 1×1 padding and convolved using 2×2 stride. Based on the given data solve the following questions 5, 6 and 7:

QUESTION 5: **Type:Comprehensive**

What is the dimension of the resulting activation map?

a) $14 \times 11 \times 32$

b) $13 \times 10 \times 32$

c) $14 \times 11 \times 3$

d) $27 \times 21 \times 32$

Correct Answer: a)

Detailed Solution:

The formula is given by:

$W_{out} = \left\lfloor \frac{W_{in} - W_{kernel} + 2P}{S} \right\rfloor + 1 = \frac{27-3+2(1)}{2} + 1 = \frac{26}{2} + 1 = 13 + 1 = 14$

$H_{out} = \left\lfloor \frac{H_{in} - H_{kernel} + 2P}{S} \right\rfloor + 1 = \frac{21-3+2(1)}{2} + 1 = \frac{20}{2} + 1 = 10 + 1 = 11$

D_{out} = number of kernels = 32

QUESTION 6: **Type:Comprehensive**

Calculate the number of parameters in the convolutional layer.

Correct Answer: 896

Detailed Solution:

Number of parameters in convolution layer = (input channel $\times$ kernel height $\times$ kernel width $\times$ no. of kernels) + no. of kernels(bias). Thus, it will be $3 \times 3 \times 3 \times 32 + 32 = 4864$

QUESTION 7: **Type:Comprehensive**

Assume that the convolutional layer is followed by a MaxPool layer with kernel size 3×3 and a stride of 1. Calculate the number of learnable parameters in this pooling layer.

Correct Answer: 0

Detailed Solution:

Number of parameters in a pooling layer is always 0.

QUESTION 8: **Type: MCQ**

A convolutional neural network is being trained on a 3-class image classification problem (Classes: Cat, Dog, Bird). For a specific training image of a Dog, the standard hard one-hot encoded ground truth label vector is $\mathbf{y} = [0, 1, 0]$.To regularize the model, we use Label Smoothing (LS), which modifies the target labels using the formula:

$$y_{i,\text{smoothed}} = (1 - \alpha) \cdot y_i + \frac{\alpha}{K}$$

Where α is the smoothing factor (0.1 in this case) and K is the total number of classes (3). Calculate the final smoothed target vector $\mathbf{y}_{\text{smoothed}}$ for this sample.

a) $[0.1000, 0.8000, 0.1000]$

b) $[0.0333, 0.9333, 0.0333]$

c) $[0.0500, 0.9000, 0.0500]$

d) $[0.0000, 0.9000, 0.1000]$

Correct Answer: b)

Detailed Solution:

$y_{\text{Dog, smoothed}} = (1 - 0.1) \cdot 1 + \frac{0.1}{3} = 0.9 + 0.0333 = 0.9333$

$y_{\text{Cat/Bird, smoothed}} = (1 - 0.1) \cdot 0 + \frac{0.1}{3} = 0.0333$

$\mathbf{y}_{\text{smoothed}} = [0.0333, 0.9333, 0.0333]$

QUESTION 9: **Type: MCQ**

Which of the following statements are true?

a) Fast-RCNN selects lesser number of region proposals from a given image compared to RCNN.

b) In Fast-RCNN, region proposals are prepared from the convolutional feature map of a given image.

c) RCNN computes CNN features on the whole input image.

d) Region proposal network (RPN) is trained to produce region proposals directly after the last convolutional layer in Faster-RCNN.

Correct Answer: d)

Detailed Solution: In Faster-RCNN, the region proposal network (RPN) is trained to produce region proposals directly after the last convolutional layer.

QUESTION 10: **Type:Numeric**

In an object detection problem, the time taken to generate all proposals is 0.75. There are 4 number of proposals generated. What is the inference time of a Faster RCNN if the convolution time and the time taken by the fully connected layer are 2.25 and 1.75, respectively? Round off the answer to 2 places of decimals.

Correct Answer: 9.25

Detailed Solution: Inference time of Faster RCNN = $1\times$ ConvTime + NumProp $\times$ fcTime = $2.25 + 4 \times 1.75 = 9.25$,

where,

NumProp = number of proposals generated

ConvTime = convolution time

fcTime = time taken by the fully connected layer.
